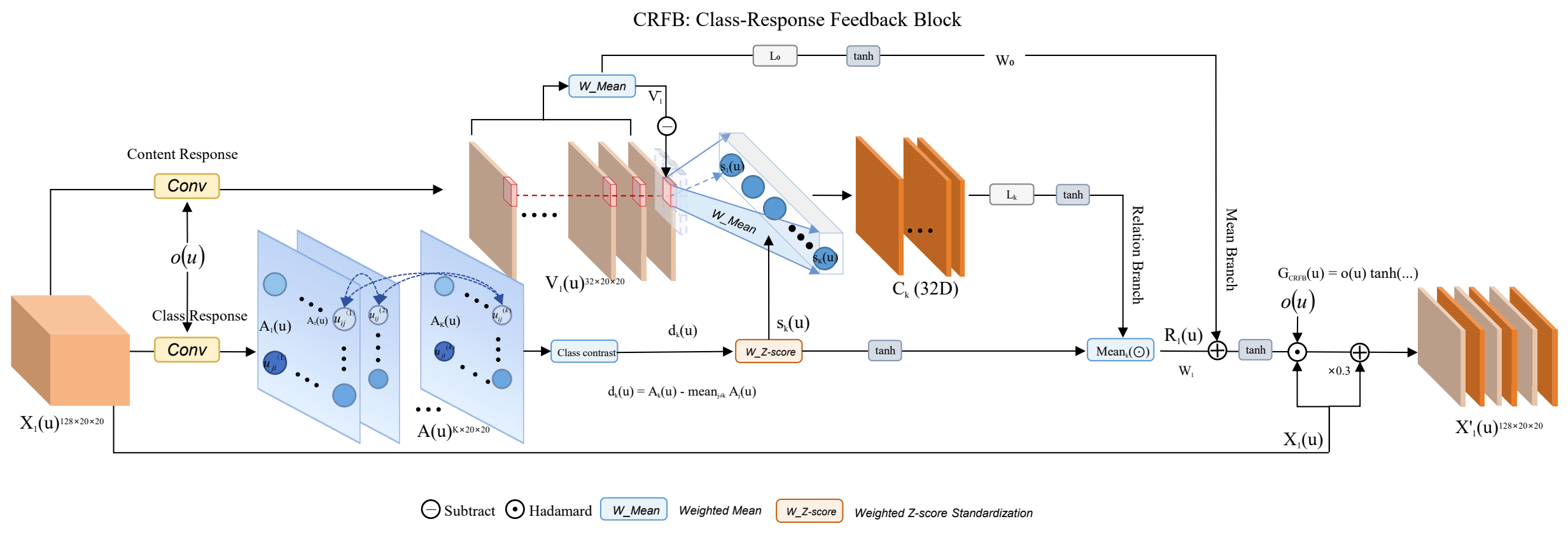

CRFB: Class-Response Feedback Block
Lo
tanh
Wo
W_Mean
V1
Content Response
Conv
o(u)
Class Response
Conv
X1(u)128×20×20
A1(u)
A2(u)
AK(u)
A(u)K×20×20
V1(u)32×20×20
s1(u)
W_Mean
sK(u)
Ck (32D)
Lk
tanh
Relation Branch
Mean Branch
dk(u)
sk(u)
Class contrast
W_Z-score
tanh
Meank(⊙)
R1(u)
W1
tanh
GCRFB(u) = o(u) tanh(...)
o(u)
×0.3
X1(u)
X'1(u)128×20×20
dk(u) = Ak(u) - meanj≠k Aj(u)
Subtract
Hadamard
W_Mean
Weighted Mean
W_Z-score
Weighted Z-score Standardization